FACULTY OF SCIENCES

Assumption-lean regression

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Motivation

- We have advocated to start a causal analysis with a choice of estimand: a model-free measure of treatment effect.

  (Hernan and Robins, 2024)

- By being model-free, debiased machine learning can be used.

  (van der Laan and Rose, 2011; Chernozhukov et al., 2018; Foster and Syrgkanis, 2024)

- Most developments for scalar estimands (e.g., $E(Y^1 - Y^0)$); some (black-box) approaches for infinite-dimensional estimands (e.g., $E(Y^1 - Y^0|X)$).
- Much progress:
  - less model misspecification bias,
  - valid data-adaptive inference,
  - simpler / more direct answers to the scientific question.

What if no concrete, uniform intervention is envisaged?

- An estimand is then difficult to choose.
  - E.g. what if all had above versus below median inflammatory markers?
  - E.g. what if pregnant mothers shifted their time of pregnancy by 2 years?
  - E.g. what if body weights of overweight people shifted to those of 'normal' weight people?
- This brings a danger of overinterpretation, extrapolation, lack of robustness and lack of insight into subject-specific effects.
- By 'losing the anchors' to a statistical modeling toolbox, well-understood statistical problems get overlooked (e.g., correlation).
- It is therefore not uncommon to return to parsimonious modeling to deliver insight that black-box methods do not always bring.

ARE WE BACK TO SQUARE ONE...?

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What are the disadvantages of fitting parsimonious models?

- modeling assumptions are used as expressing knowledge,
  rather than an approximation.
- seemingly small degrees of model misspecification can lead to large bias.

  (Rubin, 1991; Freedman, 2001; Robins and Rotnitzky, 2001; van der Laan, 2015; ...)

  - Occam's dilemma: simplicity and correctness are difficult to reconcile.

    (Breiman, 2001)

- valid data-adaptive inference is difficult to attain.
- the model often takes precedence over the scientific question,
  with difficult-to-interpret parameters that provide no immediate answer.

  (Hernan, 2010; Hernan and Robins, 2024)

CAN WE USE MODELS AS APPROXIMATIONS RATHER THAN AS ASSUMPTIONS?

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ASSUMPTION-LEAN REGRESSION

This question motivated a recent JRSS B discussion paper on assumption-lean modeling.

Vansteelandt S, Dukes O. Assumption-lean inference for generalised linear model parameters (with discussion). JRSS-B 2022

Model projection

Infer the observed data distribution using data-adaptive methods
and 'project' it onto a (semi-)parametric model.

(see also Neugebauer and van der Laan, 2007; Robins et al., 2013; Kennedy et al., 2019; Buja et al., 2019; Whitney, Shojaie and Carone, 2019; ...)

How?

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ASSUMPTION-LEAN LINEAR REGRESSION

Vansteelandt S, Dukes O. Assumption-lean inference for generalised linear model parameters (with discussion). JRSS-B 2022.

- Consider a dichotomous exposure A
  and a high-dimensional covariate L that is sufficient to adjust for confounding so that

$$E(Y^a|L) = E(Y|A = a, L).$$

- Consider a (semi-parametric) linear structural mean model of the form

$$E(Y^a|L) = \underbrace{\alpha(L)}_{\text{unknown}} + \underbrace{\beta(L)}_{\text{unknown}} a.$$

- We then **summarize the distribution of** $\beta(L)$ instead of assuming $\beta(L) = \beta$.

  (see Farrell, Liang and Misra, 2021 for an independent, related proposal)

- We will focus on the central location of $\beta(L)$.

What summary / estimand?

- We consider weighted averages

$$\frac{E\left\{w(L)\beta(L)\right\}}{E\left\{w(L)\right\}},$$

  with $\beta(L) = E(Y|A = 1, L) - E(Y|A = 0, L)$.
- Naïve interpretation as a mean difference is justified when $\beta(L) = \beta$, and not misleading otherwise.
- Weights could be fixed at $w(L) = 1$, but we recommend propensity-overlap weights

$$w(L) = \mathrm{P}(A = 1|L)\mathrm{P}(A = 0|L).$$

Why propensity-overlap weights $P(A = 1|L)P(A = 0|L)$?

- Give stability for generic use.
- Emphasize people who carry most information and are of most interest.
- Natural extension to continuous exposures.
- Because it changes the target population, we provide similarly weighted summary statistics.

Note: data are nearly always weighted, we often just don't realize (how).

Continuous exposures

- Previous proposal does not readily generalize to arbitrary exposures.
- Models for the effect of a shift in exposure

$$E\left(Y^{A+a}|A,L\right) - E\left(Y|A,L\right) = \beta a \quad \forall a,$$

  are equivalent to the partially linear model.
- If model holds, then if the exposure were shifted with a units, the mean $E\big(Y\big|A,L\big) = p$ would change to

$$E\left(Y^{A+a}|A,L\right) \approx p + \beta a$$

To ensure minimal 'squared bias'...

- ...we define β as the minimizer of

$$E\left[\int \{E(Y|A+a,L) - E(Y|A,L) - \beta a\}^2 f(A+a|L)da\right].$$

- We thus weigh shifts a by how likely the shifted exposure $A+a$ is in subjects with the same covariates.
- This prevents extrapolation.

Debiased machine learning

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Towards a model-free estimator

- The suggested minimizer equals

$$\frac{E\{\mathrm{Cov}(A, Y|L)\}}{E\{\mathrm{Var}(A|L)\}} = \frac{E[\{A - E(A|L)\}E(Y|A, L)]}{E[\{A - E(A|L)\}A]},$$

- For dichotomous exposure, it reduces to

$$\frac{E\{w(L)\beta(L)\}}{E\{w(L)\}},$$

  with $\beta(L) = E(Y|A = 1, L) - E(Y|A = 0, L)$.

- Because it is model-free, we can estimate it using machine learning.
- *How would you estimate this?*

A PLUG-IN ESTIMATOR FOR DICHOTOMOUS EXPOSURE

- First, learn $E(Y|A,L)$ and $P(A=1|L)$ via machine learning.
- Next, compute

$$\hat{\beta}(L) = \hat{E}(Y|A=1,L) - \hat{E}(Y|A=0,L)$$
$$\hat{w}(L) = \hat{P}(A=1|L)\hat{P}(A=0|L),$$

and calculate the plug-in estimator

$$\frac{\sum_{i=1}^{n} \hat{w}(L_i)\hat{\beta}(L_i)}{\sum_{i=1}^{n} \hat{w}(L_i)}.$$

A PLUG-IN ESTIMATOR FOR GENERAL EXPOSURE

- First, learn $E(Y|A, L)$ and $E(A|L)$ via machine learning.
- Next, compute
$$\hat{\beta} = \frac{\sum_{i=1}^{n} \left\{A_i - \hat{E}(A_i|L_i)\right\} \hat{E}(Y_i|A_i, L_i)}{\sum_{i=1}^{n} \left\{A_i - \hat{E}(A_i|L_i)\right\} A_i}$$
- These plug-in estimators have the same disadvantages as previously considered: they suffer from plug-in bias, and their uncertainty is difficult to assess.

How much plug-in bias?

- The bias affecting a plug-in estimator of a model-free estimand is primarily driven by minus the sample average of the estimand's efficient influence curve.

  (Pfanzagl, 1990; Bickel et al., 1993; see Hines, Dukes, Diaz-Ordaz and Vansteelandt, 2021, for a tutorial)

- An estimator that does not suffer plug-in bias sets this sample average to zero.

THE EFFICIENT INFLUENCE CURVE OF β

- The efficient influence curve of β is

$$\frac{\{A - E(A|L)\}\left[Y - E(Y|L) - \beta\{A - E(A|L)\}\right]}{E\left[\{A - E(A|L)\}^2\right]}$$

(Vansteelandt and Dukes, 2022)

- A de-biased estimator which sets its sample average to zero equals

$$\hat{\beta} = \frac{\sum_{i=1}^{n}\{A_i - \hat{E}(A_i|L_i)\}\{Y_i - \hat{E}(Y_i|L_i)\}}{\sum_{i=1}^{n}\{A_i - \hat{E}(A_i|L_i)\}^2}$$

Assumption-lean linear regression

$$E(Y^a|L) = \alpha(L) + \beta a$$

1. Predict A based on L to obtain predictions $\hat{p}$.
2. Predict Y based on L to obtain predictions $\hat{q}$.
3. Linearly regress (using least squares)

$$Y - \hat{q} \quad \text{onto} \quad A - \hat{p}$$

   with a robust standard error.

Cross-fitting is generally recommended.

Implementation 'by hand' in R

```
  # = Predict outcome = #
> model = randomForest(l,y)
> q = predict(model,l)
  # = Predict exposure = #
> modela = randomForest(l,a)
> p = predict(modela,l)
  # = Compute DML beta = #
> beta = mean((a-p)*(y-q))/mean((a-p)^2)
  # = Compute SE of DML beta = #
> sebeta = (1/sqrt(N))*sd((a-p)*(y-q-beta*(a-p)))/mean((a-p)^2)
```

Implementation 'by lm' in R

```
> library(sandwich)
> library(lmtest)
  # = Predict outcome = #
> model = randomForest(l,y)
> q = predict(model,l)
  # = Predict exposure = #
> modela = randomForest(l,a)
> p = predict(modela,l)
  # = Compute DML beta and SE = #
> model = lm(I(y-q)~-1+I(a-p))
  # = Compute SE of DML beta = #
> coeftest(model, vcov=sandwich)
```

We ask least squares software for robust / sandwich standard errors.

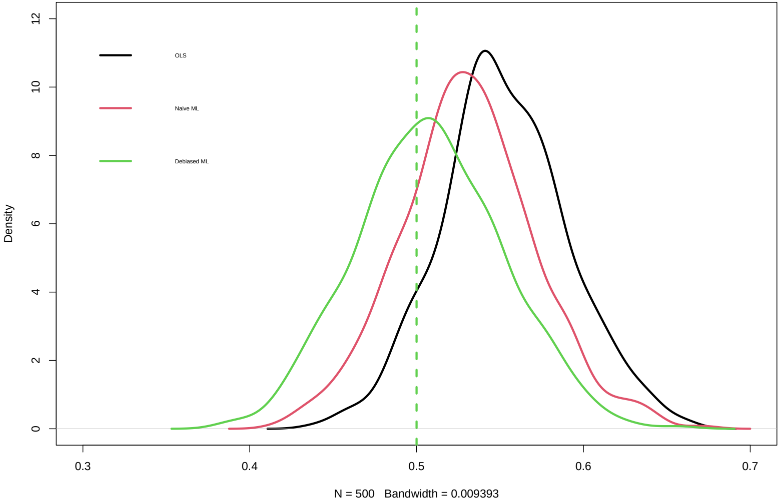
Case study: the NSW job training program

```
> re78_nsw <- lm(re78 ~ treat + age + educ + marr + nodegree + black + hisp + re74 + re75
   + u74 +u75, data = nsw_dw_cpscontrol)
> summary(re78_nsw)$coef[2,]

    Estimate   Std. Error       t value     Pr(>|t|)
1.148492e+03 5.444851e+02 2.109317e+00 3.493229e-02

> l <- as.matrix(re78_nsw$model[,-c(1,2)])
> a <- re78_nsw$model[,2]
> y <- re78_nsw$model[,1]
> sl.y <- SuperLearner(Y=y,X=data.frame(l),SL.library=c("SL.glm","SL.gam","SL.randomForest","SL.gbm"))
> sl.a <- SuperLearner(Y=a,X=data.frame(l),SL.library=c("SL.glm","SL.gam","SL.randomForest","SL.gbm"),
   family=binomial)
> q <- predict(sl.y,newdata=data.frame(l))$pred
> p <- predict(sl.a,newdata=data.frame(l))$pred
> model <- lm(I(y-q)~-1+I(a-p))
# = Compute SE of DML psi = #
> coeftest(model, vcov=sandwich)

         Estimate Std. Error t value Pr(>|t|)
I(a - p)  1345.59     636.24  2.1149  0.03445 *
```

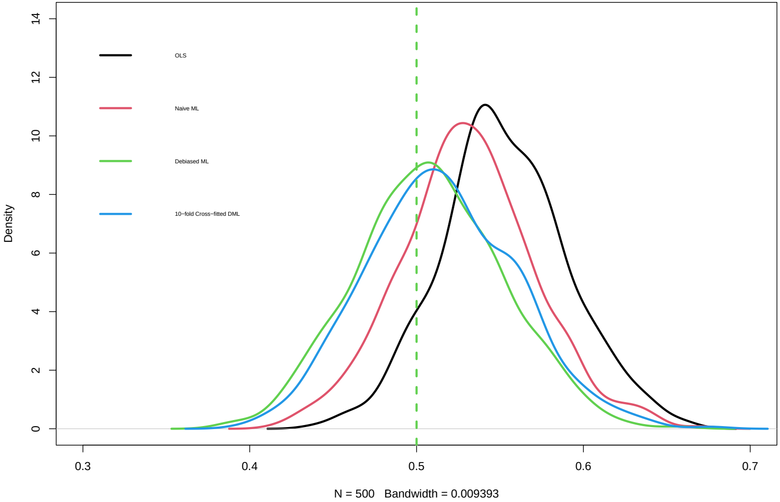
Implementation 'by hand' in R

```
> ICn = vector("numeric",length=N)
> ICd = vector("numeric",length=N)
> for (k in 1:2){
    evalset = (1+(N/2)*(k-1)):(N*k/2)
    model = randomForest(l[-evalset,],y[-evalset])
    q = predict(model,l[evalset,])
    modela = randomForest(l[-evalset,],a[-evalset])
    p = predict(modela,l[evalset,])
    V = a[evalset]-p
    ICn[evalset]=V*(y[evalset]-q)/mean(V^2)
    ICd[evalset]=V^2/mean(V^2)
}
> beta = mean(ICn)
> IC = ICn-beta*ICd
> sd(IC)/sqrt(N) # SE
```

Simulation results (1)



Simulation results (2)



Simulation results (3)

Estimator	Mean	SD	Mean SE	95% Coverage
Naïve	0.531	0.040	0.014	41.2
Debiased ML 1-fold	0.508	0.044	0.044	99.2
Debiased ML 2-fold	0.519	0.045	0.042	91.8
Debiased ML 5-fold	0.515	0.044	0.043	92.8
Debiased ML 10-fold	0.515	0.045	0.043	93.2

500 simulation runs

ZOOMING IN ON SPECIAL CASES

THERE IS MAGIC IN THE AIR...

- In case you missed it, this is real magic!

- We finally have a simple way of getting valid inference
  when our statistical analysis involves variable selection, machine learning, ...!
- I will expand on 2 specific cases to illustrate this.

Case 1: Variable selection

- Suppose our aim is to estimate β in model

$$E(Y|A, L) = \beta A + \alpha' L$$

  using variable selection to choose which components of L to retain.
- We now know how to attain valid inference.
- Compute predictions $\hat{E}(A_i|L_i)$
  using a (canonical) generalised linear model with variable selection.
- Compute predictions $\hat{E}(Y_i|L_i)$ using a linear model with variable selection.
- Compute $\hat{\beta}$ by linearly regressing $Y_i - \hat{E}(Y_i|L_i)$ on $A_i - \hat{E}(A_i|L_i)$
  using least squares software (in a model with no intercept) and sandwich standard error.
- Note that the proposal needs a different analysis for each exposure A!
  - This overcomes concerns about the 'multiplicity of models'.

Why does it work?

- Intuitively, the reason why this works is as follows.
- Variables L that are strongly associated with A, and moderately with Y,
  tend to cause much bias,
  but are difficult to detect in a model for the outcome.
- Because we now also model the exposure,
  we give ourselves 2 chances to detect them.
- The only variables that will not be detected are
  those with a very weak association with both exposure and outcome.
- But these only cause negligible bias.
- Theory requires both predictions to converge to the truth
  for the estimator and standard error to be unbiased.

Double selection

- The suggested strategy is closely related to double selection.

  (Belloni, Chernozhukov and Hansen, 2014)

- This is a specific proposal for variable selection.
- Its recommendation is to adjust for all variables found to be associated with either the exposure or the outcome (in a model that also includes the exposure).

RANDOMISED EXPERIMENTS

- Suppose that $A \perp\!\!\!\perp L$ (as often the case in randomized experiments).
- Then $\text{Var}(A|L)$ is constant so that the estimand reduces to

$$E\left\{\beta(L)\right\}$$

- For a dichotomous exposure, this is

$$E\left\{E(Y|A=1,L)-E(Y|A=0,L)\right\}=E(Y|A=1)-E(Y|A=0)$$

  regardless of whether the model is correct.
- In that case, the procedure does not require predicting A, since $E(A|L)=E(A)$ can be substituted with its sample average.
- Since these predictions are guaranteed to converge to the truth, sandwich SEs are then guaranteed valid, even when the outcome model is misspecified.

(in fact, standard OLS inference can then be shown to be valid despite variable selection (when sandwich standard errors are used).

Case 2: Risk differences

- For a dichotomous outcome Y, it is common to do logistic regression.
- Because the interpretation of coefficients as log odds ratios is subtle,
  there may be a greater interest in considering risk differences.
- It may be tempting to consider fitting a linear model,
  but this is rarely done in practice
  because such models are almost guaranteed to be misspecified.
- *Why?*
- The proposed strategy overcomes this by explicitly allowing model misspecification.

REFLECTION

Reflection (1)

- We could alternatively have chosen to fit model

$$E(Y^a|L) = \alpha' L \quad + \quad \beta a,$$

or

$$E(Y^a|L) = \alpha(L) \quad + \quad \beta a,$$

but choose to not do this:

- If this assumption is wrong, we have no clue what we are getting: *does it even capture the association between A and Y, given L?*
- Estimation in such generalized partially linear models has been studied, but is computationally demanding, cannot handle high-dimensional L well, and suffers the aforementioned problems.

What does the OLS estimator deliver when the model is wrong?

- For a dichotomous exposure A, it delivers

$$\frac{E\left[P(A=1|L)\left\{1-\tilde{\pi}(L)\right\}\left\{E(Y|A=1,L)-E(Y|A=0,L)\right\}\right]}{E\left[P(A=1|L)\left\{1-\tilde{\pi}(L)\right\}\right]} + \frac{E\left[\left\{P(A=1|L)-\tilde{\pi}(L)\right\}E(Y|A=0,L)\right]}{E\left[P(A=1|L)\left\{1-\tilde{\pi}(L)\right\}\right]},$$

  where $\tilde{\pi}(L)$ denotes the population least squares projection of A onto L.
- This is disturbing: it is complicated
  and no longer even summarising the conditional association between Y and A (given L).
- It may even differ from 0 when $A \perp\!\!\!\perp Y|L$.
- In misspecified non-linear models, we usually don't even understand what we are getting!

Reflection (2)

- Data-adaptive model building procedures explore different choices for $\alpha(L)$ in

$$E(Y^a|L) = \alpha(L) \quad + \quad \beta a.$$

- For most of these choices, the model will be misspecified.
- As we change $\alpha(L)$, we also change the magnitude of β.
- This is what complicates data-adaptive inference.

ASSUMPTION-LEAN GENERALIZED LINEAR REGRESSION

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Generalized linear models

- Model
$$E(Y|A, L) = \alpha(L) + \beta A$$
may sometimes give a poor description
of the (conditional) association between A and Y (given L).
- Especially when that association is non-linear or varies by levels of L.
- In that case, it may be useful to describe the effect on other scales.
- We will consider model
$$g\left\{E(Y^a|L)\right\} = \alpha(L) + \beta(L)a, \quad \forall a$$
with $g(.)$ known (e.g., $\log(.)$ or logit(.)).
- Estimation in such generalized partially linear models has been studied,
but is computationally demanding, cannot handle high-dimensional L well,
and suffers the aforementioned problems.

Assumption-lean generalized linear regression

Vansteelandt S, Dukes O. Assumption-lean inference for generalised linear model parameters (with discussion). JRSS-B 2022.

- For a dichotomous exposure A, we consider weighted averages

$$\frac{E\left\{w(L)\beta(L)\right\}}{E\left\{w(L)\right\}},$$

with $\beta(L) = g\left\{E(Y|A=1,L)\right\} - g\left\{E(Y|A=0,L)\right\}$.

- Naïve interpretation as e.g. a log relative risk or log odds ratio is justified when $\beta(L) = \beta$, and not misleading otherwise.
- For continuous exposures, we consider models for the effect of a shift in exposure

$$g\left\{E\left(Y^{A+a}|A,L\right)\right\} - g\left\{E\left(Y|A,L\right)\right\} = \beta a \quad \forall a,$$

which are equivalent to the generalized partially linear model.

The influence curve of β

- To ensure minimal 'squared bias', we define β as the minimizer of

$$E\left\{\int \left[g\left\{E(Y|A+a,L)\right\} - g\left\{E(Y|A,L)\right\} - \beta a\right]^2 f(A+a|L)da\right\}.$$

- For general $g(.)$, it can be shown that the influence curve of β is

$$E\left[\{A-E(A|L)\}^2\right]^{-1}\{A-E(A|L)\}\left(g\{E(Y|A,L)\} - E\left[g\{E(Y|A,L)\}|L\right]\right.$$
$$\left.+g'\{E(Y|A,L)\}\{Y-E(Y|A,L)\} - \beta\{A-E(A|L)\}\right]\Big)$$

(Vansteelandt and Dukes, 2022)

Assumption-lean logistic regression

$$\text{logit}\{E(Y^a|L)\} = \alpha(L) + \beta a$$

1. Predict A based on L to obtain predictions $\hat{p}$.
2. Predict Y based on A and L to obtain predictions $\hat{Y}$.
3. Predict $\text{logit}\left(\hat{Y}\right)$ based on L to obtain predictions $\hat{q}$.
4. Linearly regress (using least squares)

$$\text{logit}\left(\hat{Y}\right) - \hat{q} + \frac{Y - \hat{Y}}{\hat{Y}(1-\hat{Y})} \quad \text{onto} \quad A - \hat{p}$$

with a robust standard error.

Cross-fitting is generally recommended.

Assumption-lean loglinear regression

$$\log\{E(Y^a|L)\} = \alpha(L) + \beta a$$

1. Predict A based on L to obtain predictions $\hat{p}$.
2. Predict Y based on A and L to obtain predictions $\hat{Y}$.
3. Predict $\log(\hat{Y})$ based on L to obtain predictions $\hat{q}$.
4. Linearly regress (using least squares)

$$\log(\hat{Y}) - \hat{q} + \frac{Y - \hat{Y}}{\hat{Y}} \quad \text{onto} \quad A - \hat{p}$$

with a robust standard error.

Cross-fitting is generally recommended.